

18 Two protons are 1.0×10^{-14} m apart.

Approximately how many times is the electrostatic force between them greater than the gravitational force between them?

A 10^{23}

B 10^{30}

C 10^{36}

D 10^{42}

C

$$\text{gravitational force} = \frac{(6.67 \times 10^{-11})(1.67 \times 10^{-27})^2}{(1.0 \times 10^{-14})^2} = 1.86 \times 10^{-36} \text{ N}$$

$$\text{electrostatic force} = \frac{(1.60 \times 10^{-19})^2}{4\pi(8.85 \times 10^{-12})(1.0 \times 10^{-14})^2} = 2.30 \text{ N}$$